

Project Initialization and Planning Phase

Date	05 July 2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve agricultural productivity using Machine Learning by recommending the most suitable crop based on soil nutrients and environmental conditions. The system helps farmers make accurate farming decisions, improve crop yield, reduce resource wastage, and promote sustainable agriculture.

Project Overview	
Objective	The primary objective of this project is to develop a Machine Learning based system that predicts the most suitable crop using soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall, thereby helping farmers maximize agricultural productivity.
Scope	The project focuses on analyzing agricultural data and providing accurate crop recommendations through Machine Learning. It includes data preprocessing, model training, prediction, and a user-friendly web interface developed using Flask.
Problem Statement	
Description	Farmers often struggle to select the right crop because of changing climate conditions, soil nutrient imbalance, and lack of scientific decision support, resulting in low productivity and financial losses.

	The proposed system improves crop selection accuracy, increases agricultural productivity, reduces farming risks, minimizes resource wastage, and supports sustainable farming practices..
Proposed Solution	
Approach	The system applies Machine Learning algorithms to analyze soil nutrients and weather parameters, then predicts the most suitable crop using a trained model. The prediction is displayed through a simple web application built with Flask.
Key Features	1. Machine Learning based Crop Recommendation 2. Soil Nutrient Analysis (N, P, K) 3. Weather Parameter Analysis 4. Fast and Accurate Prediction 5. Flask-based User Interface 6. Easy-to-use Web Application 7. Scalable and Cost-effective Solution

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 Processor
Memory	RAM	8 GB
Storage	Disk space f	512 GB SSD
Software		
Frameworks	Python frameworks	Flask

Libraries	ML libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, joblib
Development Environment	IDE	oogle Colab, Jupyter Notebook, VS Code (optional)
Data		
Data	Agricultural Crop Recommendation Dataset	Kaggle Dataset, CSV Format, 2200 Records